Student Project Report

Sustainable Heating and Cooling Technologies
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First Student:
Name: Constantinos Paschalis
Student ID: 24-935-827

Second Student:
Name: Severin Waelty
Student ID: 20-921-581

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Abstract

This project addresses the challenge of designing an energy efficient HP (HP) system to replace conventional electric or gas heating in commercial pasta cookers. The primary goal is to reduce energy consumption while maintaining operational performance in a high-throughput restaurant environment. Our approach simulates pasta cooking dynamics and heat demand, designs a suitable subcritical HP process, and optimises key parameters such as compressor diameter and refrigerant type. Key assumptions include 0-D thermal modelling and steady environmental conditions. The final design of the system is evaluated based on energy use, startup behaviour, cycling, and control logic. Compared to a traditional electric heater, the proposed HP system significantly reduces energy use and emissions.

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1 General System Design and Constraints

1.1 Fixed Input Variables:

Fixed Input Parameters:

Variable	Value	Description
$R_{\text{env,water}}$	30 K/kW	Thermal resistance between the environment and water
$R_{\text{water,pasta}}$	15 K/kW	Thermal resistance between the water and pasta
$c_{\text{p,water}}$	4.18 kJ/(kg·K)	Heat capacity of water
$c_{\text{p,pasta}}$	3.5 kJ/(kg·K	Heat capacity of pasta
$T_{\text{pasta,start}}$	7 °C	Temperature of pasta before cooking
$T_{\text{pasta,final}}$	85 °C	Temperature of pasta after cooking
T_{env}	20 °C	Temperature of the environment
$T_{\text{water,morning}}$	20 °C	Water temperature in the morning
$T_{\text{water,min}}$	85 °C	Minimum water temperature during cooking
$T_{\text{water,max}}$	93 °C	Maximum water temperature during cooking
h_{basin}	0.2 m	Height of the whole basin
w_{basin}	0.27 m	Width of the whole basin
l_{basin}	0.47 m	Length of the whole basin
h_{cooker}	0.15 m	Height of one cooking basket
w_{cooker}	0.1 m	Width of one cooking basket
l_{cooker}	0.2 m	Length of one cooking basket
w_{wall}	0.002 m	Thickness of the basin wall
m_{pasta}	2 kg	Total dry mass of pasta per batch
t_{cooking}	420 s	Cooking time per batch
t_{break}	300 s	Break time between two batches
t_{d}	43'200 s	Full day of operation (12 h)
ρ_{w}	1000 kg/m^3	Density of water
$d_{\text{w}}^{\text{cond}}$	0.0008 m	Wall thickness of the condenser
$d_{\text{out}}^{\text{cond}}$	0.00635 m	Outer diameter of the condenser
$d_{\text{in}}^{\text{cond}}$	0.00475 m	Inner diameter of the condenser ($d_{\text{out}}^{\text{cond}} - 2 \cdot d_{\text{w}}^{\text{cond}}$)
λ_{steel}	25 W/(m·K	Thermal conductivity of Steel
$h_{\text{ref,VL}}$	2000 W/(m^2·K)	Heat transfer coefficient of the refrigerant for the vapor/liquid phase
$h_{\text{ref,V}}$	200 W/(m^2·K)	Heat transfer coefficient of the refrigerant for the vapor phase
$h_{\text{ref,L}}$	1000 W/(m^2·K)	Heat transfer coefficient of the refrigerant for the liquid phase
$h_{\text{water,L}}$	1000 W/(m^2·K)	Heat transfer coefficient of water for the liquid phase

Table 1: Fixed input parameters

1.2 Pasta Cooking Simulation and Heating Requirement Estimation

Assumptions and Restrictions

- In the morning, the water is at the temperature $T_{\text{water,morning}}$ and gets heated to $T_{\text{water,max}}$.
- Before cooking, the pasta is kept in the fridge at $T_{\text{pasta,start}}$.
- The pasta is cooked in batches of m_{pasta}, each for t_{cooking}.
- The pasta has to reach $T_{\text{pasta,final}}$ at the end of the cooking time.
- Between batches of pasta, there is a break of t_{break}.
- The water temperature must be kept between $T_{\text{water,min}}$ and $T_{\text{water,max}}$ during operation.
- The water in the water basin and the pasta can be modelled as 0-D.
- The temperature and pressure dependence of the properties of water and pasta can be neglected.
- Any evaporation of the cooking water and the water taken up by the pasta is neglected.
- At the end of the break, the water has to have a temperature of $T_{\text{water,max}}$, so every cycle is equal to each other.

Cooking Phase Dynamics
The temperature updates for pasta and water during the cooking phase are computed using discrete-time energy balance equations:

$$\forall t \in \{T_{\text{start}}^{\text{cooking}}, \ldots, T_{\text{start}}^{\text{cooking}} + t_{\text{cooking}}\}$$

$$T_{\text{pasta}}^{(t+1)} = \left(\frac{T_{\text{water}}^{(t)} - T_{\text{pasta}}^{(t)}}{R_{\text{water,pasta}} \cdot m_{\text{pasta}} \cdot c_{p,\text{pasta}}} \right) \cdot \Delta t + T_{\text{pasta}}^{(t)} \tag{1}$$

$$T_{\text{water}}^{(t+1)} = \left(\frac{-\left(\frac{T_{\text{water}}^{(t)} - T_{\text{pasta}}^{(t)}}{R_{\text{water,pasta}}}\right) - \left(\frac{T_{\text{water}}^{(t)} - T_{\text{env}}}{R_{\text{env,water}}}\right) + \dot{Q}_{\text{condenser}}}{m_{\text{water}} \cdot c_{p,\text{water}}} \right) \cdot \Delta t + T_{\text{water}}^{(t)} \tag{2}$$

Break Phase Dynamics
During the break (noncooking) period, the water is reheated using Equation 3:

$$\forall t \in \{T_{\text{start}}^{\text{break}}, \ldots, T_{\text{start}}^{\text{break}} + t_{\text{break}}\}$$

$$T_{\text{water}}^{(t+1)} = \left(\frac{-\left(\frac{T_{\text{water}}^{(t)} - T_{\text{env}}}{R_{\text{env,water}}}\right) + \dot{Q}_{\text{condenser}}}{m_{\text{water}} \cdot c_{p,\text{water}}} \right) \cdot \Delta t + T^{(t)} \tag{3}$$

Objective Function (Cooking Accuracy)

$$\varepsilon = \left| T_{\text{pasta}}^{\text{end}} - T_{\text{pasta,final}} \right| \tag{4}$$

1.3 Condenser Heat Transfer Area Limit and maximum Water Mass

Condenser Heat Transfer Area Limit:
To determine the maximum possible heat transfer area in the condenser heat exchanger, multiple assumptions had to be made.

- Heat exchanger pipes can go below, to the side, and between the baskets
- Distance between pipes is the same as the outer diameter of the pipe
- Conservative assumption: only the length of the straight part of the pipes is considered, heat transfer in turns of the pipe is not included
- The top of the basket is 0.003 m below the water surface

For all segments of the cooker – below the baskets, on the sides of the basket, and between the baskets – the calculation followed the following pattern:

1. Distance between wall or floor of the basin and the baskets or between the baskets
2. Amount of possible rows of tubes in the space. One row is equal to one plane of pipes over the whole area looked at in this step
3. Number of tubes in one row, respecting the distance between tubes
4. Length of tubes in this part of the cooker, respecting the space needed for turns

The sum of all parts in the basin resulted in a maximum heat exchanger pipe length of 83.8956 m.

Limit of Water Mass
The maximum possible water mass was calculated through the volume of the basin, subtracting the volume of the heat exchanger and the volume of the pasta, with the conversion of dry pasta to volume based on the converter [1]. Using the density of 1000 kg/m^3, the limit of the water mass was calculated based on Equation 5 and resulted in 18.023 kg.

$$m_{tot}^{water} = (v_{basin} - v_{max}^{he} - v_{pasta}) \cdot 1000\, kg/m^3 \tag{5}$$

1.4 On/Off Control Strategy

The strategy for determining the best HP operation and thermal capacity is based on a prediction strategy. For each time step, the model calculates if the HP has to be turned on, so the water temperature does not drop below 85 °C or above 93 °C. It does so by calculating the temperatures for each time step for the rest of the cycle.

This strategy leads to a temperature profile of the water and the HP operation, which is shown in Figure 1. The value defined here for heating capacity would be the optimal case, only based on

the thermodynamic equations 1, 2 and 3. Whether this is feasible in a HP cycle is part of a later step, but this defines the benchmark value.

This first optimisation results in an optimal heat capacity of 3 kW. As shown in Figure 1, the HP runs throughout the cycle and never has to be turned off.

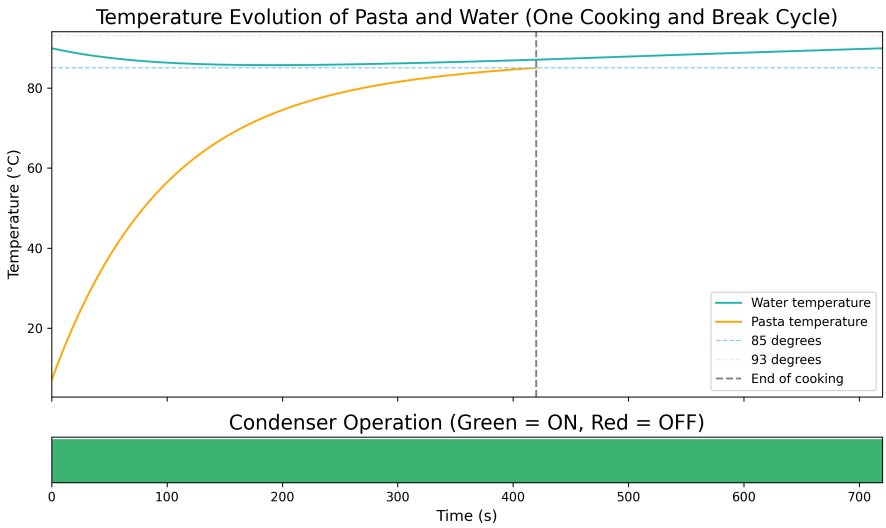

Figure 1: Temperature of water and pasta and HP operation for one optimal cooking and break cycle, not regarding the HP cycle.

2 Design of the Heat Pump Process

2.1 Assumptions, Variables, and Objective Function

Assumptions:

- Subcritical HP process
- Heat source is ambient air at 20 °C which doesn't change temperature
- The evaporator heat exchanger has no size constraint
- The heat transfer coefficients are not refrigerant specific and independent of temperature
- min_pinch $= 0.5$ [K]
- max_pinch $= 2.0$ [K]

Decision Variables:

$$\begin{aligned}
\text{Possible Refrigerants: ref} &\in \{\text{Isobutane, Butane, Isobutene, DME}\} \\
\text{Compressor cylinder diameter: D} &\in \{35,\ 40,\ 45,\ 50\} \quad [\text{mm}] \\
\Delta T_{\text{sh}} &\in \{5, \ldots, 15\} \quad [°\text{C}] \quad \text{(Superheating)} \\
\Delta T_{\text{sc}} &\in \{5, \ldots, 15\} \quad [°\text{C}] \quad \text{(Subcooling)} \\
T_{\text{co}} &\in [T_{\text{water,min}} + \text{min_pinch} + \Delta T_{\text{sc}}, \ldots, T_{\text{water,min}} + \text{max_pinch} + \Delta T_{\text{sc}}] \\
T_{\text{ev}} &\in [T_{\text{env}} - \text{max_pinch} - \Delta T_{\text{sh}}, \ldots, T_{\text{env}} - \text{min_pinch} - \Delta T_{\text{sh}}]
\end{aligned}$$

Objective function:

$$max\left(COP + x_{\text{cond}} \cdot \frac{1}{|Q_{\text{cond}} - Q_{\text{cond,opt}}|} + x_{\text{pinch}} \cdot \frac{1}{min_{pinch.difference}}\right) \tag{6}$$

Where:

COP	: Coefficient of Performance of the HP, as computed in Equation 7
Q_{cond}	: Maximum feasible condenser heat output
$Q_{\text{cond,opt}}$	: Optimal condenser heat load from unconstrained simulation
x_{cond}	: Weighting factor for condenser deviation penalty
x_{pinch}	: Weighting factor for pinch constraint penalty
$\min(\text{pinch difference})$	: Minimum pinch across all heat exchange sections

$$COP_{\text{inner}} = \frac{q_{\text{high}}}{w_{\text{comp}}} = \frac{h_{\text{state,2}} - h_{\text{state,3}}}{h_{\text{state,2}} - h_{\text{state,1}}} \tag{7}$$

Where:

q_{high}	: specific heat released in the condenser in [kJ/kg]
w_{comp}	: specific work of the compressor in [kJ/kg]
$h_{\text{state,X}}$	: specific enthalpy at state X in the thermodynamic cycle

Due to losses and the on/off strategy, the HP must produce more heat than the pasta actually needs to reach the 85 °C. To account for this circumstance, the COP has to be corrected, which is done as shown in Equation 8.

$$COP_{\text{res}} = COP_{\text{inner}}\left(\frac{\frac{Q_{\text{pasta}}}{Q_{\text{HP}}}}{0.9\frac{Q_{\text{pasta}}}{Q_{\text{HP}}} + 0.1}\right) \tag{8}$$

Where:

Q_{pasta}	: Heat needed to cook the pasta in [kJ/kg]
Q_{HP}	: Total Heat produced by the HP in [kJ/kg]

2.2 Optimisation of the Heat Pump Process

All functions mentioned here are based on those used in the exercises of the ETH course *Sustainable Heating and Cooling Technologies (151-0950-00L)*. Some have been changed slightly and can be studied in more detail in the Jupyter notebook file, described in Section 8. Important to mention is that the optimisation function – shown in Equation 6 –optimises mainly for the condenser power because we want to maximise the COP_{res}, which would decrease with turning the heat pump on and off multiple times. The COP is almost negligible when the condenser power approaches the optimum calculated in Section 1.4.

Optimisation Procedure:

1. Calculate the isentropic efficiency and mass flow rate, based on the unchanged function *recip_comp_SP*, given in the project description.
2. Simulate the cycle, based on *Fluid_CP*, and get the temperature difference at the evaporator inlet and outlet.
3. Simulate the cycle, based on *Fluid_CP* and get the temperature difference at the condenser inlet and outlet, as well as at the dew point.

4. Get the minimum pinch out of all these temperature differences if it meets the constraint of being between min_pinch and max_pinch.
5. Calculate COP, heat output, and sink temperatures. Continue if the new heat output is larger than the previously calculated optimum of 3 kW, because it cannot be any better than that.
6. Get the condenser length, calculated using the function *condenser_len_calc*, and check if it is smaller than the maximum length of the heat exchanger pipe of 83.8956 m.
7. Compute the object function and save the new values if it corresponds to a new optimum.

2.3 New On/Off Control Strategy

Based on the now optimal configuration of the HP process, the entire day was simulated. In a first step, the optimal start temperature was determined to begin a cooking cycle using the prediction approach described in Section 1.4. Knowing this temperature, the time to heat the water in the morning can be calculated. The rest of the day was then simulated, making sure that the pasta reaches the desired temperature and the temperature at the end of a break is again equal to the optimal cooking start temperature.

3 Selection of the Most Suitable Combination

3.1 Comparison of Refrigerants and the resulting Best Configuration

The optimised variables for all refrigerants except *DME* are shown in Table 2. The model was unable to find a HP configuration for the *DME* refrigerant that satisfies all constraints. The selected refrigerant was *Butane* because it was closest to the optimal condenser power. The resulting thermodynamic cycle of the best configuration is shown in Figure 7.

ref [-]	T_{ev} [°C]	ΔT_{sh} [°C]	T_{co} [°C]	ΔT_{sc} [°C]	Q_{cond} [kW]	L_{cond} [m]	D [mm]	η_{is} [-]	COP_{inner} [-]	batches [#]
Isobutane	4.0	14	103.0	17	3.0007	33.4	35	0.57	1.95	unknown
Butane	**5**	**14**	**100.5**	**15**	**3.00007**	**36.1**	**40**	**0.53**	**2.005**	55
Isobutene	5.5	13	100.5	15	3.00016	36.1	35	0.57	2.09	unknown

Table 2: Heat pump parameters for the optimized HP cycle

3.2 Comparison of different Parameters

Minimum Pinch
One design parameter was the minimum pinch. It is desirable to keep it small because it leads to better heat extraction and rejection and better second-law efficiency (exergy).

As the heat source temperature was fixed and the heat sink temperature was computed in the optimisation, the minimum pinch was dependent on the evaporation and condensation temperatures. The change in minimum pinch for these two temperatures is shown in Figure 2. The model determined that an evaporation temperature of 5 °C and a condensation temperature of 100.5 °C to be optimal, see Table 2.

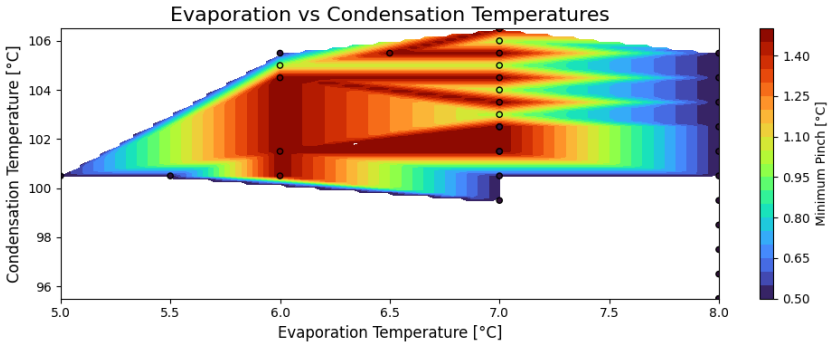

Figure 2: Change in minimum pinch of the whole cycle for different evaporation and condensation temperatures.

Condenser Power

The most important parameter in this project is the condenser power, because it is the variable the model optimises for. The focus was laid on the condenser power instead of the COP because this model was based on a perfect foresight assumption which allowed to get the perfect condenser power to run the HP continuously. The condenser power is influenced by many design parameters. The obvious ones are condenser-specific, e.g. condensation temperature and subcooling. The way in which changing these parameters influences the condensation power is shown in Figure 5. With respect to all other constraints and decision variables, the model was optimised for a subcooling temperature of 15 °C – see Table 2. In addition, a well-designed compressor is important. The change in power output based on the compressor parameters cylinder diameter and isentropic efficiencies is shown in Figure 3. The best configuration based on the model turned out to be a diameter of 40 mm and an isentropic efficiency of 0.53 – see Table 2. The only design part that is missing from the HP is now the evaporator. The way in which the evaporation and superheating temperatures play together is displayed in Figure 4. The optimiser chose a superheating of 14 °C – see Table 2.

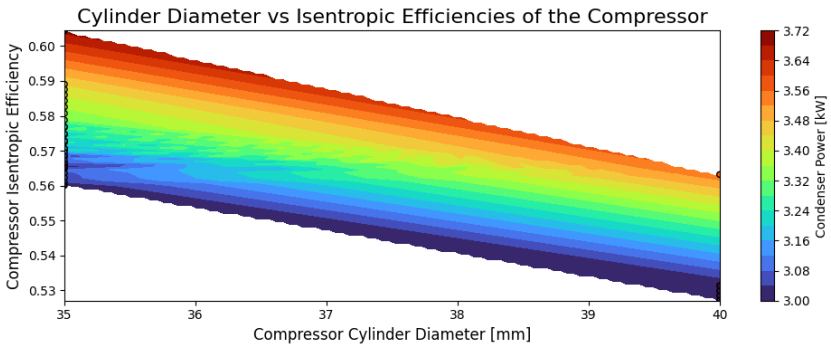

Figure 3: Change in condenser power for different cylinder diameters and isentropic efficiencies of the compressor.

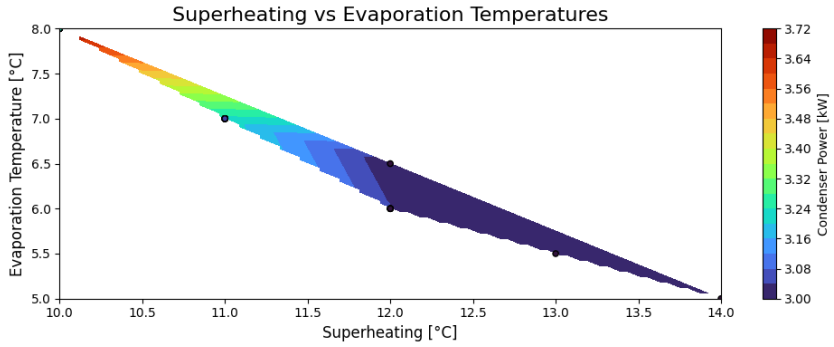

Figure 4: Change in condenser power for different evaporation and superheating temperatures.

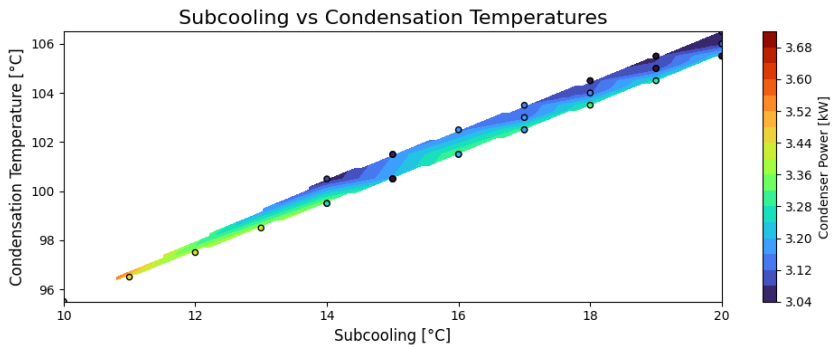

Figure 5: Change in condenser power for different condensation and subcooling temperatures.

COP

In Figure 6 the change in COP_{inner} is displayed for different combinations of the mass flow rate of the refrigerant and the condenser power. The two variables are linearly correlated, which is expected. The optimiser chose a mass flow rate of 7.65 g/s. Due to the decision that the COP was not the main objective variable, it is clearly shown that the optimiser chose a low COP. This can be explained that the resulting COP after correction – as computed in Equation 8 – was only computed after the whole optimisation. Because the strategy here followed was to ensure that the HP is always running, the COP does not fluctuate because the HP is always turned on and off, and the reduction in COP is kept low to obtain a COP_{res} of 1.77.

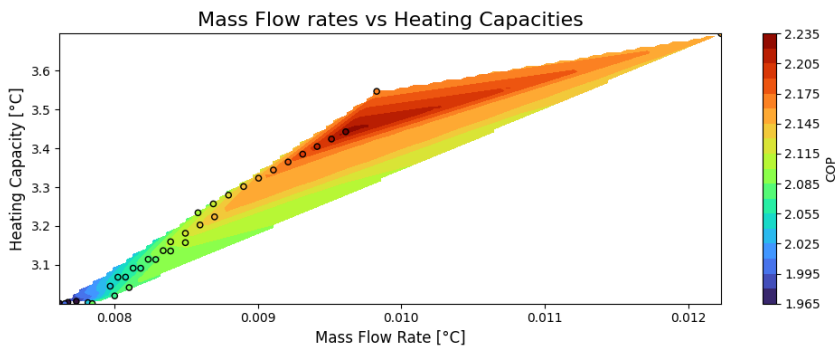

Figure 6: Change in COP for different mass flow rates and condenser powers.

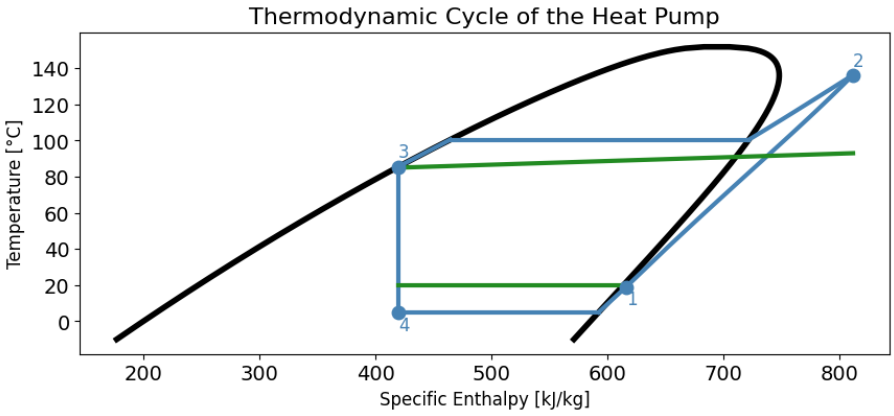

Figure 7: Optimised thermodynamic cycle

3.3 System Visualization

In Figure 8 it is shown that when the first people enter the restaurant and start the cooker, the system needs just under an hour to get the water to the perfect starting temperature to cook a batch of pasta. This is enough time to prepare everything else in the kitchen and restaurant, but not too long so that you would have to wait for the water to heat up when you would already be ready. **Morning Heat Up**

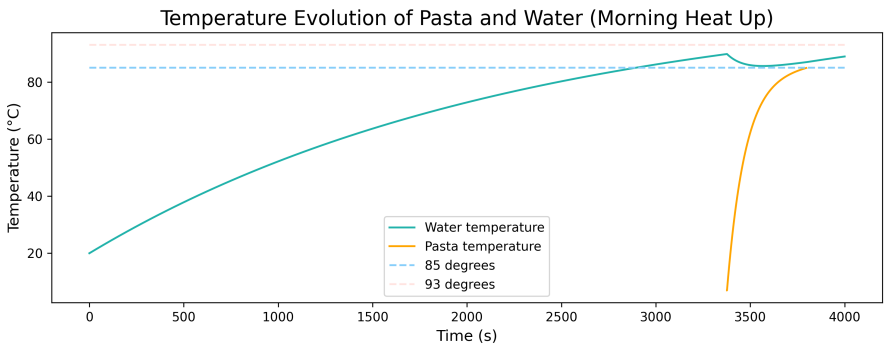

Figure 8: Shows the evolution of the water temperature in the morning until it reaches the optimal cooking start time temperature.

One full Cycle
A complete cooking cycle is shown in Figure 9 and shows the condenser operation time below the graph of water and pasta temperatures. Compared to the best-case scenario shown in Figure 1, the differences are marginal. This displays, that the model was able to choose a HP configuration that reached a condenser power that was almost optimal.

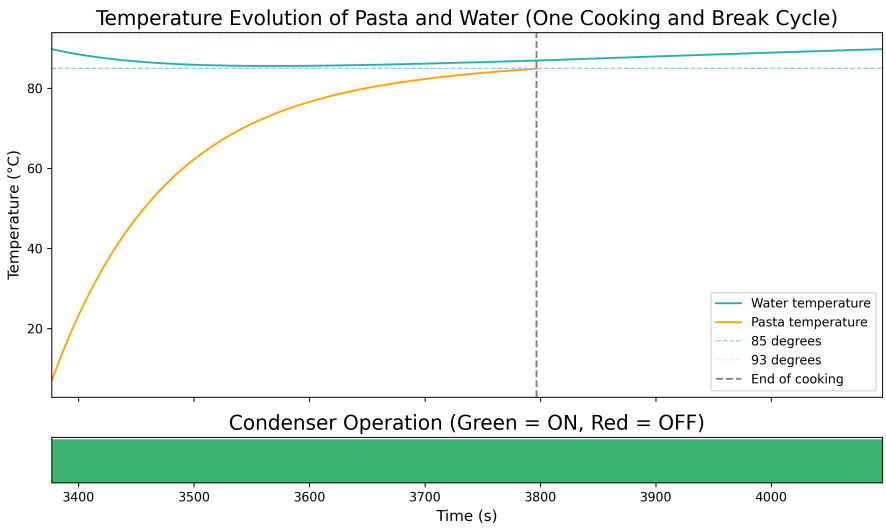

Figure 9: This plot displays the water and pasta temperatures as well as the condenser operation of one full cooking and break cycle.

Full day of operation

The evolution of the water and pasta temperatures throughout the day of operation is shown in Figure 10. With this configuration, the restaurant can cook 55 batches of pasta (2), which corresponds to 220 portions. Every cooking cycle has the exact same temperature profile, which ensures a perfect pasta temperature and quality at the end of the cooking time.

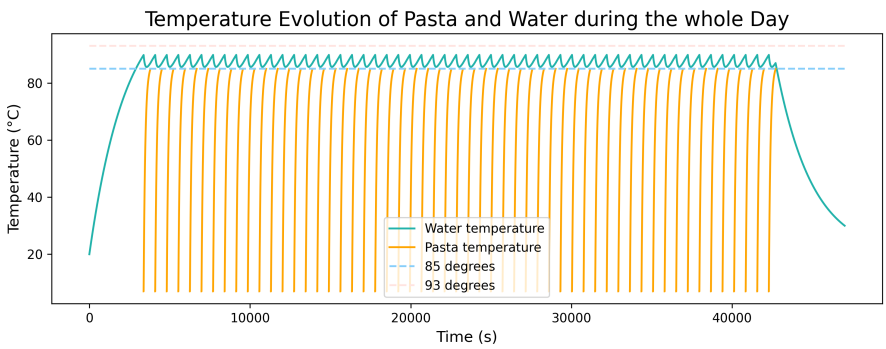

Figure 10: Displays the water and pasta temperatures for a full 12 h day of operation.

Evening Cool Down

In Figure 11 the last cooking cycle and the following cooling of the water after turning off the HP are shown. The important thing to mention is that the model automatically stops the HP operation after the last pasta batch if it knows that there will not be enough time left for one more break and cooking cycle. The water then cools and will be at 20 °C again in the next morning.

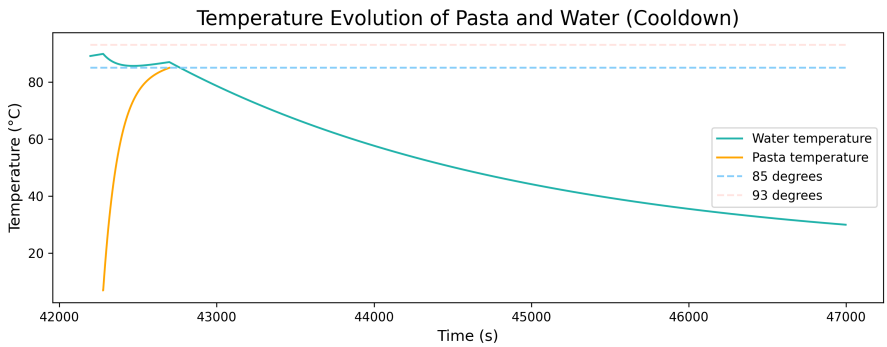

Figure 11: Cooldown of the water at the end of the day.

4 Comparison to an Electric Heater (EH) and Final Recommendation

4.1 Energy, Cost and CO2 Emissions Analysis

The comparison in energy usage, electricity cost and CO_2 emissions between the optimised HP system and an electric heater is shown in Figure 12. Because the HP is running non-stop on full capacity, the computation of the three values was very straightforward. The exact computation is shown in Equations 9, 10, and 11, where the only difference between the HP and the electric heater lies in the COP. The HP COP was calculated as described in Equation 8 and the COP of the electric heater was assumed to be 1, according to the US Department of Energy [4]. Therefore, the factor between the two technologies in the three values is equal to the COP_{inner} of the HP.

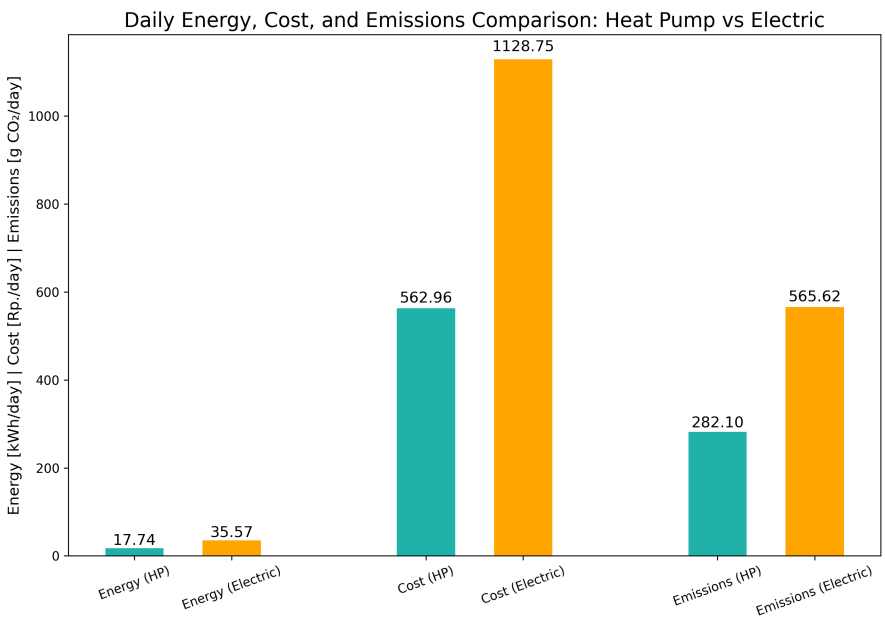

Figure 12: Comparison of energy usage, electricity cost and CO_2 emissions of an electric heater and the designed HP system.

$$E_{\mathsf{X}} = \frac{t_{\mathsf{on}} \cdot Q_{\mathsf{X}}}{COP_{\mathsf{X}}} \tag{9}$$

$$Cost_{\mathsf{X}} = E_{\mathsf{X}} \cdot p_{\mathsf{el}} \tag{10}$$

$$Emissions_{\mathsf{X}} = E_{\mathsf{X}} \cdot x_{\mathsf{CO_2}} \tag{11}$$

Where:

$X \in \{HP, EH\}$	: Technologies
E_{X}	: Technology specific electricity consumption
t_{on}	: Runtime for one day
Q_{cond}	: Heat output for one day
COP_{X}	: Technology specific COP
p_{el}	: Electricity price of Zurich – ewz [2]
$x_{\mathsf{CO_2}}$	: Emissions Factor of the electricity mix of ewz [3]

4.2 Final Recommendation

Based on the results of the optimisation and simulation, the designed heat pump system described in detail in Section 3 is the recommended configuration. It meets the daily heat demand of a commercial pasta cooker with stable and continuous operation and delivers a corrected COP of 1.77. Compared to a traditional electric heater, the system achieves a significant reduction in electricity use, operational cost, and CO_2 emissions. Given the available space for the heat exchanger and the assumed ambient conditions, the design is both feasible and efficient. Therefore, we recommend the implementation of this heat pump concept as a sustainable and economically attractive alternative to resistive electric heating in high-throughput kitchen environments.

5 Uncertainty Analysis

This project is based on several assumptions that introduce uncertainties in the results. One main simplification is the use of 0-D thermal models for both water and pasta, meaning temperature is assumed to be uniform throughout. In reality, there would be temperature gradients, especially during heating, that could affect the accuracy of the heat transfer calculations. Also, thermal properties such as heat capacities are assumed constant and independent of temperature, although they can vary slightly in real systems.

Some effects are not included in the model, such as the reduction in water volume due to pasta that absorbs water during cooking. This could affect the total heat capacity of the system and slightly reduce the heating demand over time, or if you fill it up after each cycle, increase the heat demand. The ambient air temperature used as the heat source is assumed to be constant at 20 °C, which is unlikely in a real kitchen and could lead to optimistic estimates of HP performance. The heat transfer coefficients are also simplified and not specific to the refrigerant used, which may affect the comparison between the refrigerants and the predicted COP values.

The most important uncertainties are the simplified heat transfer modelling, the neglect of water uptake by pasta, and the fixed ambient temperature. All of these factors could influence the final energy demand and system performance. Although assumptions make the model easier to work with, they should be reviewed in more detailed future studies or compensated for by adding safety margins if this model should be developed further.

6 Conclusion

In this project, a heat pump system was developed and optimised that was adapted to the thermal requirements of a commercial pasta cooking process. By combining a simplified dynamic model of cooking behaviour with a thermodynamic cycle optimisation, we identified a feasible system configuration that maintains cooking performance while reducing energy use. The analysis shows that the proposed solution reduces both electricity consumption and CO_2 emissions by a factor of 1.77 compared to an electric heater. Although based on simplified assumptions, the results provide a solid foundation for future design refinement and potential real-world application. The proposed system demonstrates that sustainable kitchen operations can be achieved through the integration of targeted heat pumps and intelligent control.

7 Bibliography

References

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[4] U.S.Department of Energy. Electric Resistance Heating, 2025. Available online: https://www.energy.gov/energysaver/electric-resistance-heating#:~:text=Electric%20resistance%20heating%20is%20100,the%20fuel's%20energy%20into%20electricity.

8 Attached Files

The attached Jupyter notebook contains all the functions and simulations used throughout the project. This includes the cooking process model, calculations of the heat balance for the cooking and break phases, and simulation of the temperature dynamics of water and pasta. Core optimisation routines evaluate different refrigerant and compressor configurations to identify the best-performing heat pump set-up. Control logic functions determine the heat pump's on/off behaviour over a full operational day. Additional scripts produce all result plots, including energy demand, cost comparison, emissions, and temperature profiles. The constants and input parameters are stored centrally for easy modification. All results and all optimisation steps are stored in a CSV file in the /Results folder. The plots used in the report are saved in the /Plots folder.

Running the Code:
Because the code was developed on Google collab, the whole project should be put on your Google Drive following the path in Cell 3 in the code. If the directory is correct, the code should run without problem and take approximately one hour to run completely.